

Desmos Mastery - Day 4

17 original questions | Calculator permitted

Complete the questions before turning to the answer explanations. Mark any guess, even if it turns out to be correct. For math, show your setup and check the requested quantity. For reading, identify the exact supporting detail; for grammar, identify the clause or subject.

These questions were authored for this workbook from the lesson topics. They are not official SAT questions or copies of the course's paid practice.

Inequalities | 10 questions

[Lesson video: Inequalities](#)

1. Which point satisfies both inequalities below?

$$y < 2x + 1$$
$$y \geq x + 2$$

- A) (2, 5)
- B) (2, 4)
- C) (3, 7)
- D) (1, 4)

2. If $x = 4$, what is the greatest integer y satisfying both inequalities?

$$y < 3x - 3$$
$$y \geq x + 2$$

Answer: _____

3. Which interval describes all real x satisfying the inequality below?

$$5 - 2x < 11$$

- A) $x < -3$
- B) $x > -3$
- C) $x < 3$
- D) $x > 3$

4. Which pair describes all solutions of the inequalities below?

$$\begin{aligned}2x + 1 &\geq 7 \\ 3x - 2 &< 19\end{aligned}$$

- A) $x \geq 3$ and $x < 7$
- B) $x > 3$ and $x \leq 7$
- C) $x \geq 3$ and $x \leq 7$
- D) $x \leq 3$ or $x \geq 7$

5. Tickets cost \$12 each, and a \$9 booking fee is charged once. With a budget of at most \$100, what is the greatest number of tickets that can be bought?

Answer: _____

6. Which graph description matches $y \geq -x + 3$?

- A) Dashed boundary; shading below
- B) Dashed boundary; shading above
- C) Solid boundary; shading above
- D) Solid boundary; shading below

7. How many integer values of x satisfy $|x - 2| < 4$?

Answer: _____

8. Which value of x satisfies both $x^2 < 16$ and $x \geq 3$?

- A) -3
- B) 2
- C) 3
- D) 4

9. At $x = 3$, how many integer y -values satisfy all three inequalities below?

$$y > x - 1$$

$$y \leq 2x$$

$$x + y < 8$$

Answer: _____

10. The feasible region is defined below. Which point lies on an included boundary and also satisfies every inequality?

$$x + y \leq 4$$

$$x > 0$$

$$y < 5$$

A) (0, 0)

B) (1, 5)

C) (2, 2)

D) (5, 0)

Circle Equations | 7 questions

[Lesson video: Circle Equations](#)

11. What is the center of the circle below?

$$(x - 3)^2 + (y + 2)^2 = 49$$

A) (3, -2)

B) (-3, 2)

C) (3, 2)

D) (-3, -2)

12. What is the radius of the circle below?

$$2x^2 - 8x + 2y^2 + 12y = 24$$

Answer: _____

13. The endpoints of a diameter are $(-4, 1)$ and $(8, 1)$. What is the x-coordinate of the center?

Answer: _____

14. Which point lies on the circle below?

$$(x - 2)^2 + (y - 1)^2 = 9$$

- A) $(5, 2)$
- B) $(2, 4)$
- C) $(1, 1)$
- D) $(6, 1)$

15. A circle has center $(-1, 4)$ and radius 6. What is the greatest y-coordinate of a point on the circle?

Answer: _____

16. For the circle below, which point is not in its interior?

$$x^2 + y^2 = 4$$

- A) $(0, 0)$
- B) $(1, 1)$
- C) $(2, 0)$
- D) $(1, 0)$

17. The circle below has radius r . What is r^2 ?

$$x^2 + 6x + y^2 - 4y = 7$$

Answer: _____

Answers & explanations

Count correct answers, but also review correct guesses. Redo missed questions after a day without looking at the solution.

1: B 2: 8 3: B 4: A 5: 7 6: C 7: 7 8: C 9: 2 10: C 11: A 12: 5 13: 2 14: B 15: 10 16: C 17: 20

Why each answer works

- 1. Answer: B.** At $(2, 4)$, $4 < 5$ and $4 \geq 4$. A lies on the excluded first boundary, C fails the first inequality, and D fails it as well.
- 2. Answer: 8.** The inequalities become $y < 9$ and $y \geq 6$. The greatest integer is 8.
- 3. Answer: B.** Subtract 5: $-2x < 6$. Divide by -2 and reverse the inequality to obtain $x > -3$.
- 4. Answer: A.** The first gives $x \geq 3$. The second gives $x < 7$. Both must hold, including 3 and excluding 7.
- 5. Answer: 7.** $12n + 9 \leq 100$ gives $n \leq 91/12 \approx 7.58$. The ticket count must be an integer, so at most 7.
- 6. Answer: C.** Equality is included, so the line is solid. Greater y -values lie above the line.
- 7. Answer: 7.** The inequality is $-4 < x - 2 < 4$, or $-2 < x < 6$. The integers are $-1, 0, 1, 2, 3, 4$, and 5 .
- 8. Answer: C.** The quadratic inequality requires $-4 < x < 4$. Intersecting with $x \geq 3$ gives $3 \leq x < 4$. Only 3 is offered.
- 9. Answer: 2.** Substitute $x = 3$: $y > 2$, $y \leq 6$, and $y < 5$. The integers are 3 and 4.
- 10. Answer: C.** At $(2, 2)$, $x + y = 4$ is included, $x > 0$, and $y < 5$. The other points fail at least one condition.
- 11. Answer: A.** Compare $(x - h)^2 + (y - k)^2 = r^2$. Here $h = 3$ and $k = -2$.
- 12. Answer: 5.** Divide by 2: $x^2 - 4x + y^2 + 6y = 12$. Complete the squares: $(x - 2)^2 + (y + 3)^2 = 25$. Radius = 5.
- 13. Answer: 2.** The center is the midpoint: $((-4 + 8)/2, 1) = (2, 1)$.
- 14. Answer: B.** The center is $(2, 1)$ and radius 3. $(2, 4)$ is 3 units away. The squared distances of the other points are 10, 1, and 16.
- 15. Answer: 10.** The highest point is 6 units above the center: $4 + 6 = 10$.
- 16. Answer: C.** The point $(2, 0)$ satisfies $x^2 + y^2 = 4$ and lies on the boundary. A point not in the interior can be on the boundary or outside.
- 17. Answer: 20.** Complete the squares: $(x + 3)^2 + (y - 2)^2 = 7 + 9 + 4 = 20$. The question asks for r^2 , so do not take the square root.

For each missed or guessed question, record the skill, the condition you overlooked, and one specific action for next time.